BASEMENT LEVEL FIRE ALARM LAYOUT

FIRST LEVEL FIRE ALARM LAYOUT

Jason Kahan

Digitally signed by Jason Kahan
DN: C=US, E=jk@jasonkahan.com, CN=Jason Kahan
Date: 2025.10.08 08:16:46-04'00'

COMMONWEALTH OF MASSACHUSETTS
JASON CATALFO KAHAN
FIRE PROTECTION
NO. 48388
REGISTERED
PROFESSIONAL ENGINEER

10/08/2025

NOTES

NB DESIGN LLC
109 TENNYSON DR
MANCHESTER,NH 03104

REVISIONS		
DATE	DESCRIPTION	BY

SPRINKLER SYMBOLS

FIRE ALARM
RESIDENTIAL SYSTEM

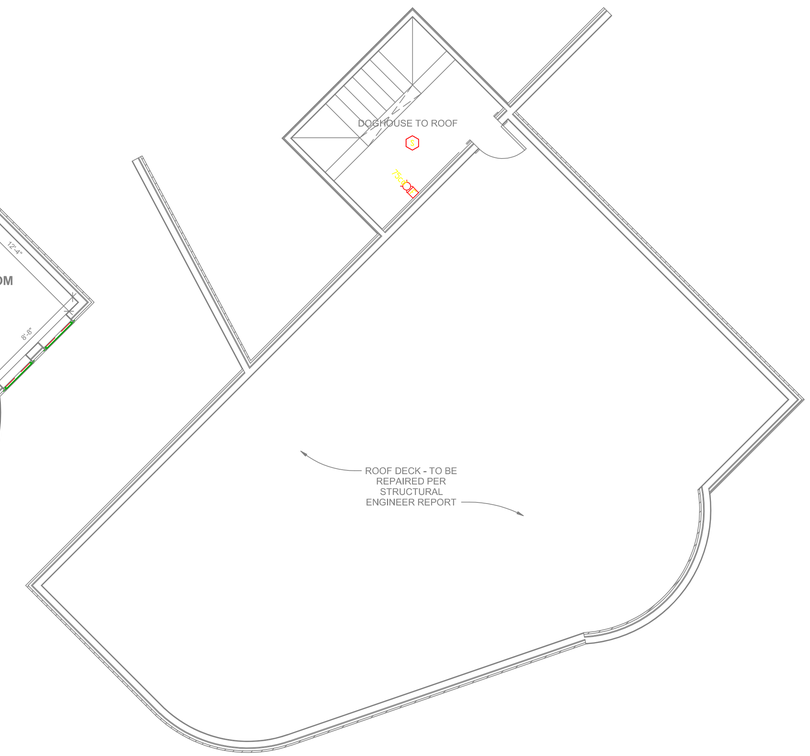
RESIDENTIAL MULTI UNIT
252 ASPINWALL AVE
BROOKLINE, MA 02445

CONTRACTOR:

FIRE ALARM BASE-1ST	
PERMIT NO.	
CONTRACT NO.	
APPROVAL	
DRAWN BY	NB
SCALE	1/4" = 1'-0"
DATE	10/6/25
REVISED	
PLOTTED	FA 1

SECOND LEVEL FIRE ALARM LAYOUT

THIRD LEVEL FIRE ALARM LAYOUT



ROOF LEVEL FIRE ALARM LAYOUT

COMMONWEALTH OF MASSACHUSETTS
JASON CATALFO
KAHAN FIRE PROTECTION
NO. 46398
REGISTERED PROFESSIONAL ENGINEER
10/08/2025

NOTES

NB DESIGN LLC
109 TENNYSON DR
MANCHESTER,NH 03104

REVISIONS		
DATE	DESCRIPTION	BY

SPRINKLER SYMBOLS

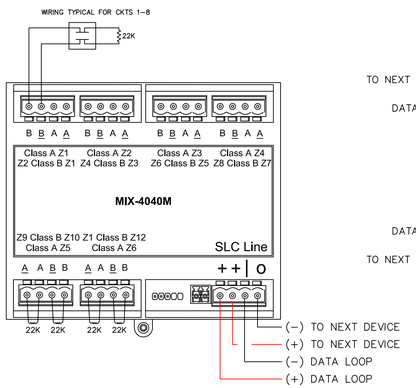
FIRE ALARM
RESIDENTIAL SYSTEM

RESIDENTIAL MULTI UNIT
252 ASPINWALL AVE
BROOKLINE, MA 02445

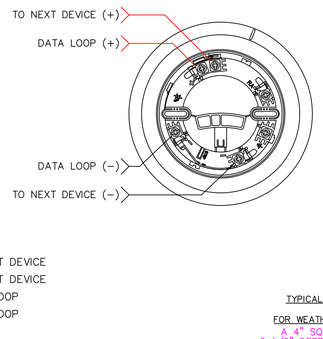
CONTRACTOR:

FIRE ALARM 2ND-ROOF	
PERMIT NO.	
CONTRACT NO.	
APPROVAL	
DRAWN BY	NB
SCALE	3/16" = 1'-0"
DATE	10/6/25
REVISED	FA 2
PLOTTED	

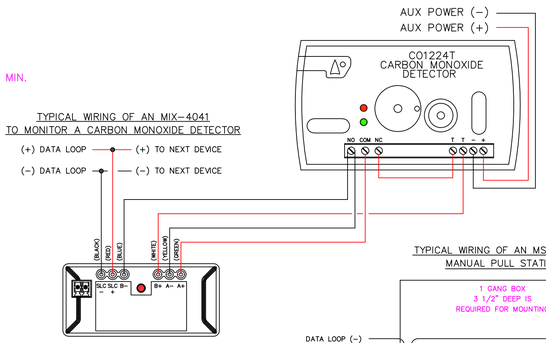
TYPICAL WIRING OF AN MIX-4040M
MULTI INPUT MODULE
(6 CLASS A/12 CLASS B)



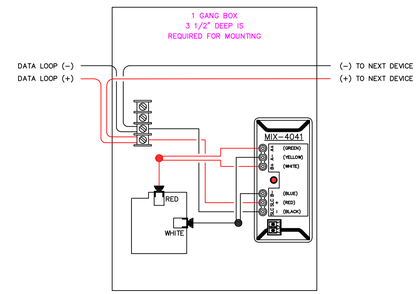
TYPICAL WIRING OF
MIX-4001 DETECTOR BASES
A 4" SQUARE BOX W/WO PLASTER RING,
4 1/4" OCTAGON OR 1 GANG BOX 2 1/8" DEEP MIN.
IS REQUIRED FOR MOUNTING



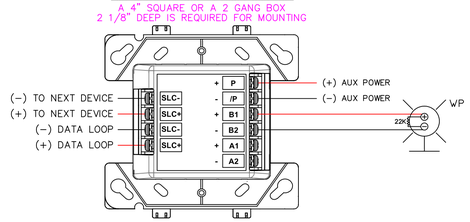
TYPICAL WIRING OF AN MIX-4041
TO MONITOR A CARBON MONOXIDE DETECTOR



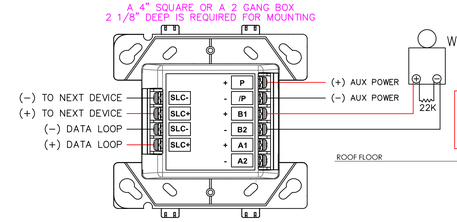
TYPICAL WIRING OF AN MS-710MPU
MANUAL PULL STATION



TYPICAL WIRING OF AN MIX-4046
SIGNAL MODULE
FOR WEATHERPROOF BEACON CONTROL
A 4" SQUARE OR A 2 GANG BOX
2 1/8" DEEP IS REQUIRED FOR MOUNTING



TYPICAL WIRING OF AN MIX-4046
SIGNAL MODULE
FOR SPRINKLER BELL CONTROL
A 4" SQUARE OR A 2 GANG BOX
2 1/8" DEEP IS REQUIRED FOR MOUNTING



FIRE ALARM LEGEND

NOTIFICATION

- 75cd FIRE ALARM SYSTEM ADA TYPE 90db HORN/ STROBE UNIT. SEE MOUNTING DETAIL FOR HEIGHTS. 75cd = INDICATES CANDELA RATING (AS SHOWN ON DRAWINGS)
- 15cd FIRE ALARM SYSTEM ADA TYPE 135db STROBE ONLY UNIT. SEE MOUNTING DETAIL FOR HEIGHTS. 15cd = INDICATES CANDELA RATING (AS SHOWN ON DRAWINGS)
- 90db FIRE ALARM SYSTEM ADA TYPE 90db LOW FREQUENCY (520HZ) TYPE MINI HORN UNIT. SEE MOUNTING DETAIL FOR HEIGHTS.
- FIRE ALARM SYSTEM WEATHER PROOF SPRINKLER FLOW ALARM BELL. SEE MOUNTING DETAIL FOR HEIGHTS.

INITIATING

- MANUAL PULL STATION, MOUNTED 48" ABOVE FINISHED FLOOR
- CEILING MOUNTED PHOTOELECTRIC. SYSTEM TYPE SMOKE DETECTOR
- AUTOMATIC HEAT DETECTOR 135 DEGREES FIXED TEMPERATURE WITH ZONE ADDRESSABLE MODULE.
- CEILING MOUNTED PHOTOELECTRIC, SYSTEM CARBON MONOXIDE (CO) DETECTOR.
- LOCAL 120 VOLT, TANDEM WIRED PHOTOELECTRIC SMOKE DETECTOR WITH INTEGRAL BATTERY BACKUP
- LOCAL 120 VOLT, TANDEM WIRED COMBINATION OF PHOTOELECTRIC SMOKE/CARBON MONOXIDE DETECTOR WITH INTEGRAL BATTERY BACKUP
- ADDRESSABLE MONITOR MODULE.

EQUIPMENT

- FACP — FIRE ALARM CONTROL PANEL, "FACP"
- FIRE ALARM ANNUNCIATOR, "FAA"
- FIRE ALARM WEATHERPROOF BEACON - FLASHING TYPE (NOT ROTATING)
- 24/80 HOUR BATTERY
- IP/CELLULAR COMMUNICATOR
- FIRE PROTECTION SYSTEM FLOW SWITCH, FURNISHED AND INSTALLED BY THE SPRINKLER CONTRACTOR, WIRED BY ELECTRICAL CONTRACTOR.
- FIRE PROTECTION SYSTEM TAMPER SWITCH, FURNISHED AND INSTALLED BY THE SPRINKLER CONTRACTOR, WIRED BY ELECTRICAL CONTRACTOR.

WIRING

- 2/C #14 POWER LIMITED FIRE ALARM CABLE TYPE "FPLP" (INITIATING)
- 2/C #14 POWER LIMITED FIRE ALARM CABLE TYPE "FPLP" (NOTIFICATION)
- DENOTES WATERPROOF

NOTES

1. No AC voltage on any terminal other than those specified.
2. Maximum line resistance is 40 Ohms, maximum line capacitance is 0.5 micro-farad per loop.
3. This illustration is shown in proximity to the FACP for connectivity purposes only. It is strongly recommended to place this surge suppressor in the electrical cabinet supplying the branch circuit feeding the FACP.

NOTICE

Ground terminal must be connected to Earth Ground per Article 760 of the National Electric Code.
Failure to make a proper Earth Ground connection to a metal cold water pipe or a driven Ground Rod to this terminal will result in loss of Lightning protection, reduce the tolerance of the system to transients and will adversely affect its operation.
Panel Neutral or Conduit Ground is not acceptable.
Minimum Wire Size: 12 AWG

WARNING

THE INSTALLING CONTRACTOR IS RESPONSIBLE FOR THE WIRING OF ALL DEVICES.
THE WIRE TYPE AND THAT ALL WIRE SIZES SHALL MEET STATE AND LOCAL CODE REQUIREMENTS.
THE INSTALLING CONTRACTOR IS ALSO RESPONSIBLE THAT THE INSTALLATION IS FREE OF ALL DEFECTS.

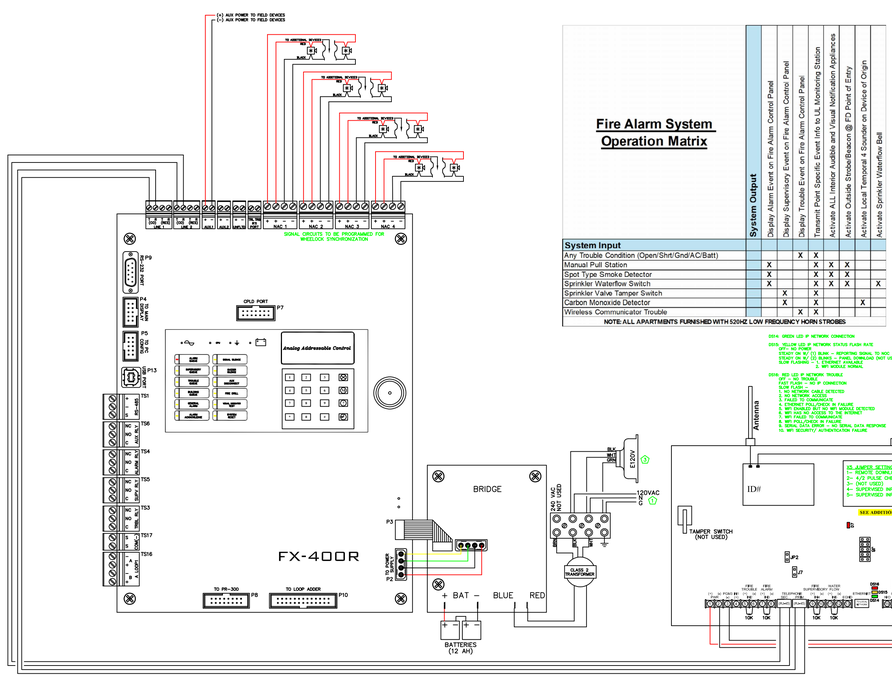
WIRING SPECIFICATIONS

- DATA/SLC- 16 AWG TWISTED CABLE. SHIELDED CABLE NOT ACCEPTED.
- ZONE/IDC- 16 AWG.
- SIGNAL/NAC- 16AWG MIN 12AWG MAX
- DC POWER- 14AWG
- AC POWER- 12AWG. POWER LIMITED AND NON POWER LIMITED CABLE CAN NOT BE RUN THROUGH THE SAME RACEWAY.

GENERAL NOTES

1. ALL Fire Alarm System wiring shall comply with the National Electrical Code, Applicable State, and Local Fire and Safety Codes. As well as being coordinated with the Local Authority having Jurisdiction.
2. CAUTION:DO NOT connect any power to control panel (batteries or 120vac) until all other field wiring is tested.
3. DO NOT Install F.A.C.P or smoke detectors in any unheated areas. For all other devices refer to manufactures published instructions.
4. DO NOT Install any A.C. current carrying conductors close to or in the same raceway with fire alarm system conductors.
5. Dotted Lines Indicate factory wiring. Solid lines indicate connections to be made by the installer.
6. See Installation Manual for additional wiring instructions.
7. All relays are shown in normal supervisory condition. All relay dry contacts: 28 VDC @ 1A ONLY.
8. F.A.C.P. Indicating circuit outputs are: 24 VDC @ 1.7 amps per circuit.
9. All electronic devices must be installed in a "DRY" temperature controlled environment.
10. All batteries must be installed in a temperature controlled (72-82°F) environment. Failure to do so will alter battery life.
11. Cabinet Dimensions:

FX-400R - H 23.0000" W 16.7000" D 5.7800"

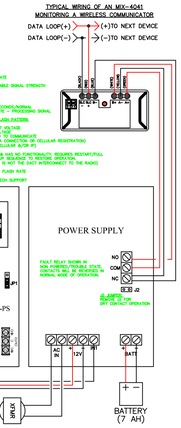


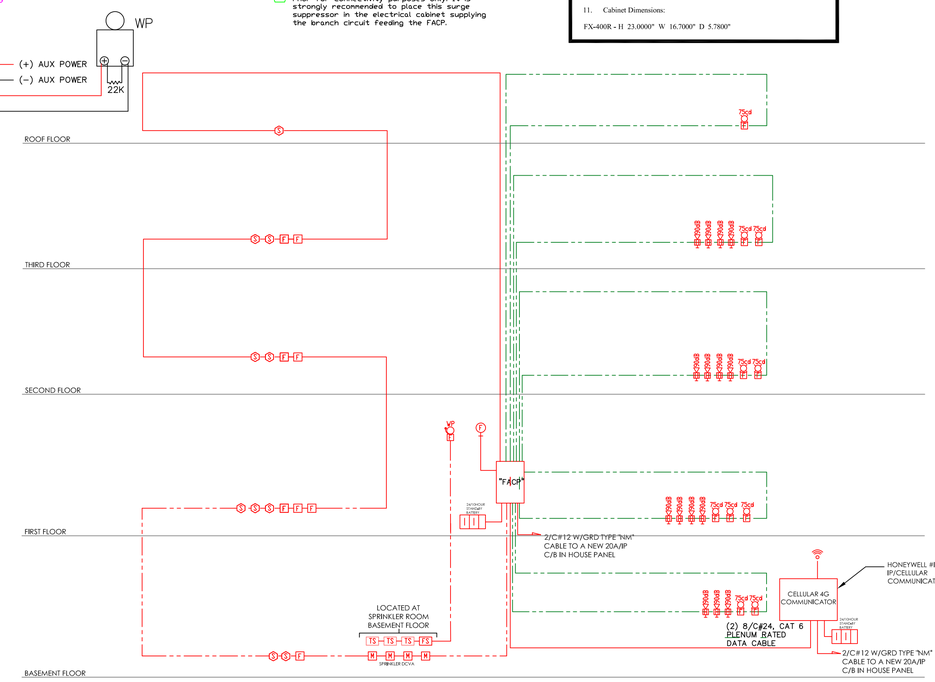
Fire Alarm System Operation Matrix

System Input	System Output	Display Alarm Event on Fire Alarm Control Panel	Display Supervisory Event on Fire Alarm Control Panel	Display Trouble Event on Fire Alarm Control Panel	Transmit Point Specific Event Info to UL Monitoring Station	Activate ALL Interior Audible and Visual Notification Appliances	Activate Outside Strobe/Beacon @ FD Point of Entry	Activate Local Temporal 4 Sounder on Device of Origin	Activate Sprinkler Waterflow Bell
Any Trouble Condition (Open/Shrt/Gnd/AC/Batt)			X	X					
Manual Pull Station		X		X	X	X	X		
Spot Type Smoke Detector		X		X	X	X	X		
Sprinkler Waterflow Switch		X		X	X	X	X		X
Sprinkler Valve Tamper Switch			X	X					
Carbon Monoxide Detector			X	X				X	
Wireless Communicator Trouble			X	X					

NOTE: ALL APARTMENTS FURNISHED WITH 520HZ LOW FREQUENCY HORN STROBES

TYPICAL WIRING OF AN MIX-4041
MONITORING A WIRELESS COMMUNICATOR





NOTES

NB DESIGN LLC
109 TENNYSON DR
MANCHESTER,NH 03104

REVISIONS

DATE	DESCRIPTION	BY

SPRINKLER SYMBOLS

COMMONWEALTH OF MASSACHUSETTS
JASON CATALFO KAHAN
FIRE PROTECTION
NO. 40381
REGISTERED PROFESSIONAL ENGINEER
10/06/2025

FIRE ALARM
RESIDENTIAL SYSTEM

RESIDENTIAL MULTI UNIT
252 ASPINWALL AVE
BROOKLINE, MA 02445

CONTRACTOR:

FIRE ALARM RISER/DETAILS

PERMIT NO.	
CONTRACT NO.	
APPROVAL	
DRAWN BY	NB
SCALE	1/4" = 1'-0"
DATE	10/6/25
REVISED	
PLOTTED	

FA 3